Tapered kaleidoscope — 100 mm prototype

Cutting and assembly

Three identical printed rails, each carrying one flat trapezoidal mirror, lapped together at the corners into a truncated triangular pyramid. Because the three mirror planes converge on a common apex 100 mm beyond the narrow end, the images tile a *sphere* rather than the flat honeycomb an untapered scope produces. You look in the wide end; the glass chips go at the narrow end.

Mirror-face triangle	25.40 mm at the narrow end, 50.80 mm at the wide end
Length along the axis	100 mm
Taper half-angle	4.1936°
Corner dihedral	60.529286° *(not 60 — the excess is the whole effect)*
Image	a sphere subtending 60°, about 453 facets
Mirror blank	trapezoid, parallel edges 25.40 and 50.80, height 100.27, legs 101.07 mm

1. Print the rails

Three copies of the same part — there is no left or right hand. About 1.5 hours each.

The model already sits with its outside face on the XY plane, so it needs no support and the exterior comes off the bed smooth. That does make the mirror pocket floor a *top* surface, so **turn on ironing for top surfaces**. With thin floppy mirror stock the mirror conforms to whatever that floor does, and print texture becomes optical texture. With rigid mirror it does not matter.

2. About the mirror stock

The cheap "unbreakable" tiles are plastic, not glass, so ignore anything you read about glass cutters and running pliers. Two honest warnings:

- **They are sold for safety, not optics, and are usually noticeably wavy.** Flatness is the single most important optical property of a kaleidoscope mirror, so expect a soft, smeary image with visible distortion in the outer facets. Do not judge the design by what these produce — judge the fit, the corners and the workflow.
- **Measure the thickness before printing** and set the feature's `mirrorThickness` to match. The default is 1.6 mm. If the pocket is wrong the corners will not close.

3. Cutting the mirrors

Do not use the rotary cutters. They are built to shear fabric against a mat, not to score rigid plastic. The blades are thin and hardened and will chip on the first pass, and they cost more than the tiles. (If the sheet turns out to be under about half a millimetre and genuinely floppy, a rotary cutter will work — test on a scrap corner.) The utility knife and the sewing ruler are the right pair; the ruler's printed grid is better for this than a plain straightedge.

1. **Check the template.** Print `mirror-template.pdf` at 100 % — *Actual size*, not *Fit to page* — and verify both rulers on it with a real ruler. If they are off, every dimension is off.
2. **Leave the protective film on** and mark on it. It prevents scratches and the blade tracks better.
3. **Transfer the marks.** Tape the template over the tile, prick through the centre of all twelve circled

marks with a pin, lift the paper, and join the pricks with the straightedge. Marking through paper beats cutting through it — a knife shreds paper and then wanders.

4. **Score.** Put the ruler on the side you are keeping, so a slip runs into the waste. Hold it down hard; more failures here come from the ruler creeping than from the knife wandering. Draw the knife along the line six to ten times with moderate, consistent pressure, same track each time, aiming to get about halfway through. Do not try to power through in one pass.

5. **Snap — or don't.** Score up, overhang past a table edge, hold the tile flat, press down sharply and evenly. Thin plastic often *bends instead of breaking*; unlike glass it has no clean cleavage. If it flexes rather than snapping, stop and keep scoring until you are through. Slower, but far more reliable, and on 100 mm pieces it is only a couple of minutes.

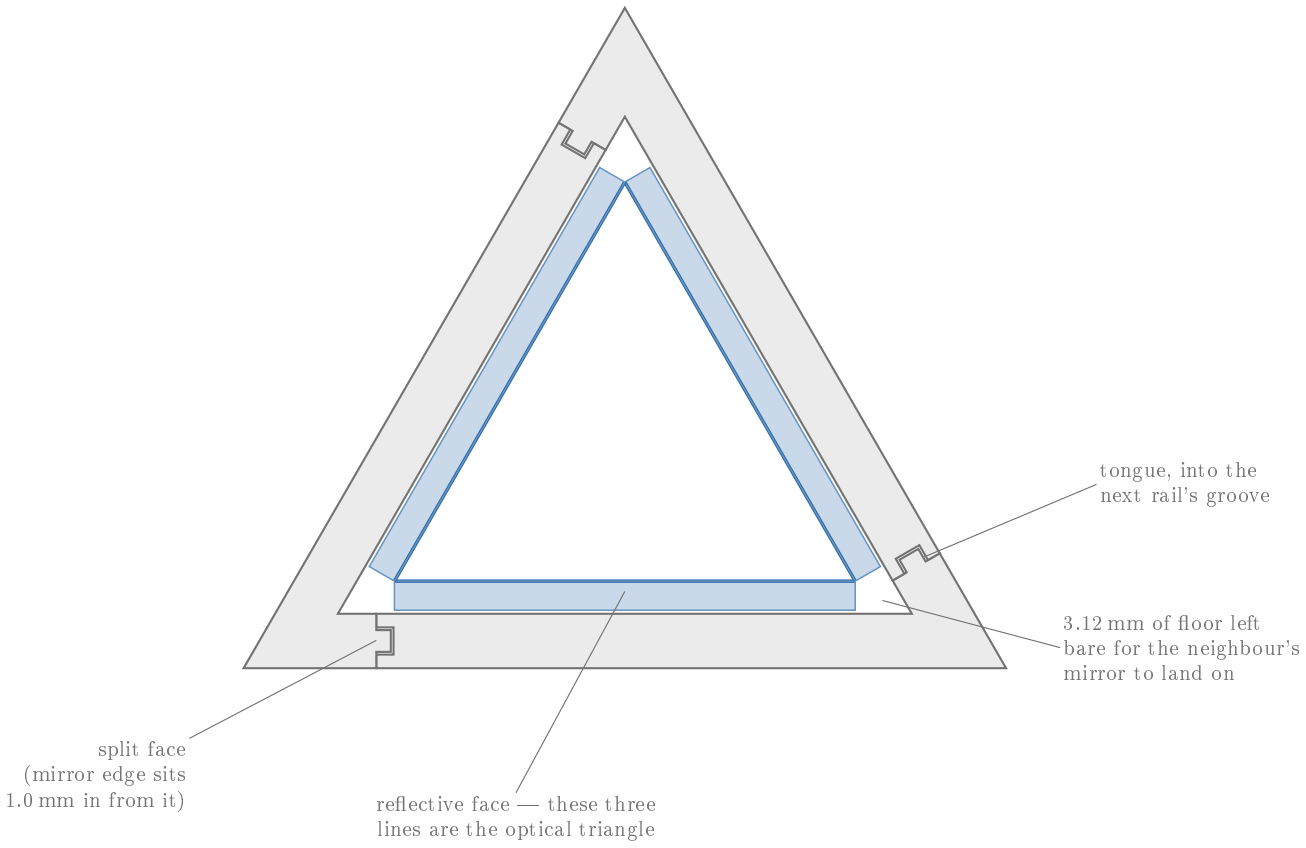
6. **Finish the two long edges.** Lay 400-grit flat on the table, hold the piece vertical, two or three light passes per edge. Keep it square — do not round it over, and do not sand onto the reflective face.

7. **Check the coating.** Hold each long edge up to the light. The silver must run right to the edge; any dead band there prints as a dark web across every facet.

Only two of the four edges matter. The slanted 101.07 mm sides butt against the neighbouring mirrors. The 25.40 and 50.80 mm ends are hidden under the end caps and can be rough. Cut on the line — these outlines are finished size and a scored edge removes essentially nothing.

4. Where the mirror sits

narrow-end cross-section, 2.4:1



The pocket floor is deliberately wider than the mirror, and asymmetrically so. On the split-plane side it runs past the mirror edge by exactly 1.0 mm; on the corner-block side it runs past by 3.12 mm, and that band is where the *neighbouring* mirror's body and bond layer land.

So the rule is simply: **set each mirror's edge 1.0 mm in from the flat split face**, ends flush with the rail ends. The other side takes care of itself.

Handedness sorts itself out. All three rails are identical, so if you bond each mirror the same way onto its own rail, the pinwheel comes out right automatically. There is no way to get one mirror backwards without noticing.

5. Dry-assemble before you glue anything

Mirrors sitting loose in the pockets, three rails pressed together, masking tape around the outside. Look through it. Thirty seconds of feedback on whether the corners close and the pattern coheres, with nothing committed.

What to look for:

- **Do the tongues seat in the grooves** and the three rails close into a triangle without a gap at the last joint?
- **Do the mirror edges meet at the vertices** without overlapping or leaving a visible gap? A dark line at every cell boundary means the mirrors are a touch narrow or the coating has pulled back from the edge.
- **Is there a sphere?** It should read as a ball floating at the far end, about 60° across, not as a flat honeycomb. If it looks flat, something is wrong with the taper, not with your cutting.

6. Bond and close up

Thin double-sided tape or a few dots of clear silicone. The model assumes a 0.2 mm bond layer, so measure your tape and set `bondGap` to match, or use silicone and squeeze it down. A uniform bond layer only resizes the image sphere very slightly; an uneven one tilts a mirror, which is the error that actually matters.

Then glue the three corner joints, band or tape the assembly while it sets, and add the object chamber and eyepiece.

Companion files: `mirror-template.pdf` (1:1 cutting template), `kaleidoscope-side.fs` (the Onshape feature, with the geometry derivations in its header), `scope-sim.html` (ray-traced preview — enter dihedral 60.5293 and ratio 2 to see what this prototype should look like).